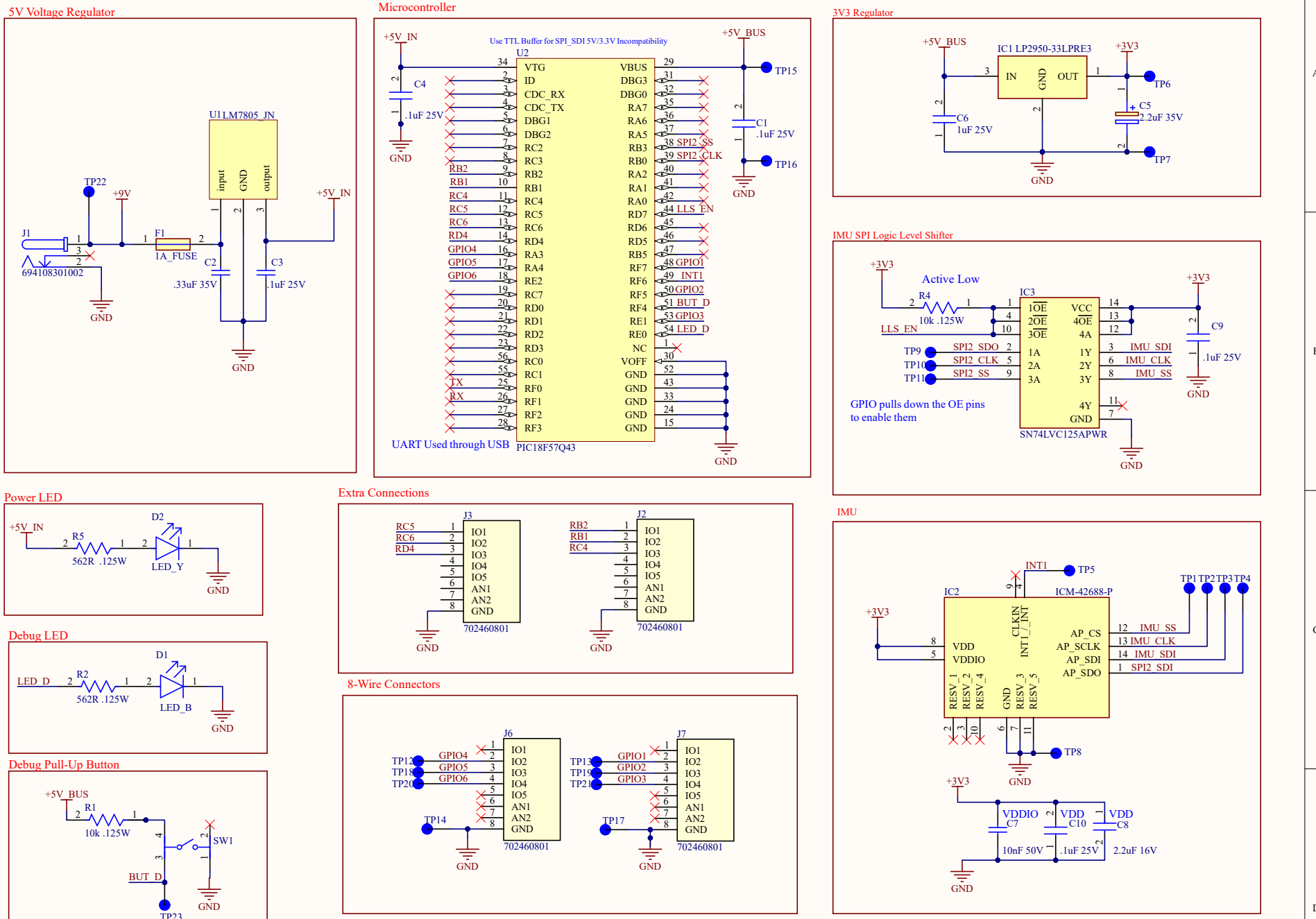

5V Voltage Regulator
U1 LM7805_IN
input
GND
output
TP22
+9V
+5V_IN
J1
694108301002
F1
1A_FUSE
C2
.33uF 35V
C3
.1uF 25V
GND
Microcontroller
Use TTL Buffer for SPI_SDI 5V/3.3V Incompatibility
U2
PIC18F57Q43
+5V_IN
+5V_BUS
C4
.1uF 25V
C1
.1uF 25V
TP15
TP16
VTG
ID
CDC_RX
CDC_TX
DBG1
DBG2
RC2
RC3
RB2
RB1
RC4
RC5
RC6
RD4
GPIO4
GPIO5
GPIO6
RC7
RD0
RD1
RD2
RD3
RC0
RC1
TX
RF0
RX
RF1
RF2
RF3
VBUS
DBG3
DBG0
RA7
RA6
RA5
RB3
RB0
RA2
RA1
RA0
RD7
RD6
RD5
RB5
RF7
RF6
RF5
RF4
RE1
RE0
NC
VOFF
SPI2_SS
SPI2_CLK
LLS_EN
GPIO1
INT1
GPIO2
BUT_D
GPIO3
LED_D
UART Used through USB
3V3 Regulator
+5V_BUS
IC1 LP2950-33LPRE3
IN
GND
OUT
+3V3
TP6
TP7
C5
2.2uF 35V
C6
1uF 25V
IMU SPI Logic Level Shifter
+3V3
Active Low
R4
10k .125W
LLS_EN
IC3
SN74LVC125APWR
1OE
2OE
3OE
4OE
VCC
1A
2A
3A
4A
1Y
2Y
3Y
4Y
TP9
TP10
TP11
SPI2_SDO
SPI2_CLK
SPI2_SS
IMU_SDI
IMU_CLK
IMU_SS
C9
.1uF 25V
GPIO pulls down the OE pins
to enable them
Power LED
D2
LED_Y
R5
562R .125W
Debug LED
D1
LED_B
R2
562R .125W
LED_D
Debug Pull-Up Button
R1
10k .125W
SW1
BUT_D
TP23
Extra Connections
J3
J2
RC5
RC6
RD4
RB2
RB1
RC4
IO1
IO2
IO3
IO4
IO5
AN1
AN2
GND
702460801
8-Wire Connectors
J6
J7
TP12
TP18
TP20
TP13
TP19
TP21
GPIO4
GPIO5
GPIO6
GPIO1
GPIO2
GPIO3
TP14
TP17
IMU
IC2
ICM-42688-P
INT1
TP5
TP1
TP2
TP3
TP4
VDD
VDDIO
CLKIN
INT1 / INT
AP_CS
AP_SCLK
AP_SDI
AP_SDO
IMU_SS
IMU_CLK
IMU_SDI
SPI2_SDI
RESV_1
RESV_2
RESV_4
RESV_3
RESV_5
TP8
VDDIO
C7
10nF 50V
VDD
C10
.1uF 25V
VDD
C8
2.2uF 16V
Title
PIC1857Q43 IMU Board
Size
A
Number
Revision
1
Date:
10/30/2026
Sheet of
File:
C:\Users\..\egr304_board.SchDoc
Drawn By:
Joseph Nunez